

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer 2024

Course: B. Tech.

Semester: VIII

Branch: Mechanical Engineering / Mechanical Engineering (Sandwich) / Production Engineering

Subject Code & Name: BTMEC801A Fundamentals of Automotive Systems

Max Marks: 60

Date: 13/06/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Explain the phenomena of knock in SI engines.	CO1	6
B) What is an engine block? Explain wet sleeve and dry sleeve inserts with neat sketches.	CO1, CO2, CO3	6
C) Define the following with respect to engine performance. a) Indicated power b) Indicated thermal efficiency c) Brake thermal efficiency d) Mechanical efficiency	CO1, CO2, CO3	6
Q.2 Solve Any Two of the following.		12
A) Discuss the various functions of a camshaft in an IC engine.	CO1, CO2, CO3	6
B) With neat sketches explain different phases of Combustion in CI Engine.	CO2	6
C) Explain in detail the emission control systems used in IC engine.	CO1	6
Q.3 Solve Any Two of the following.		12
A) What is a Clutch? Explain Construction and working of Single Plate clutch with sketch.	CO2, CO3, CO4	6
B) Explain the working of a synchromesh gear box used in Automobiles, with neat sketch.	CO2, CO3, CO4	6
C) Explain the requirements for an efficient automobile clutch.	CO2, CO3, CO4	6
Q.4 Solve Any Two of the following.		12
A) Explain with neat sketch working of Macpherson Strut Suspension system.	CO2, CO3,	6
B) Explain with neat sketch the components and working of a Drum Brake.	CO2, CO3,	6
C) What is ABS? Discuss with a neat sketch the operation Methodology of ABS.	CO2, CO3,	6
Q.5 Solve Any Two of the following.		12
A) Explain with neat sketch working of hydraulic power steering system.	CO2, CO3, CO4	6
B) Explain the terms (i) Camber angle (ii) Castor Angle (iii) Steering axis Inclination (iv) Toe-In Toe-Out.	CO2, CO3, CO4	6
C) Differentiate between Tyres with tube and Tubeless tyres.	CO2, CO3	6

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